

Chapter 2(part 1)

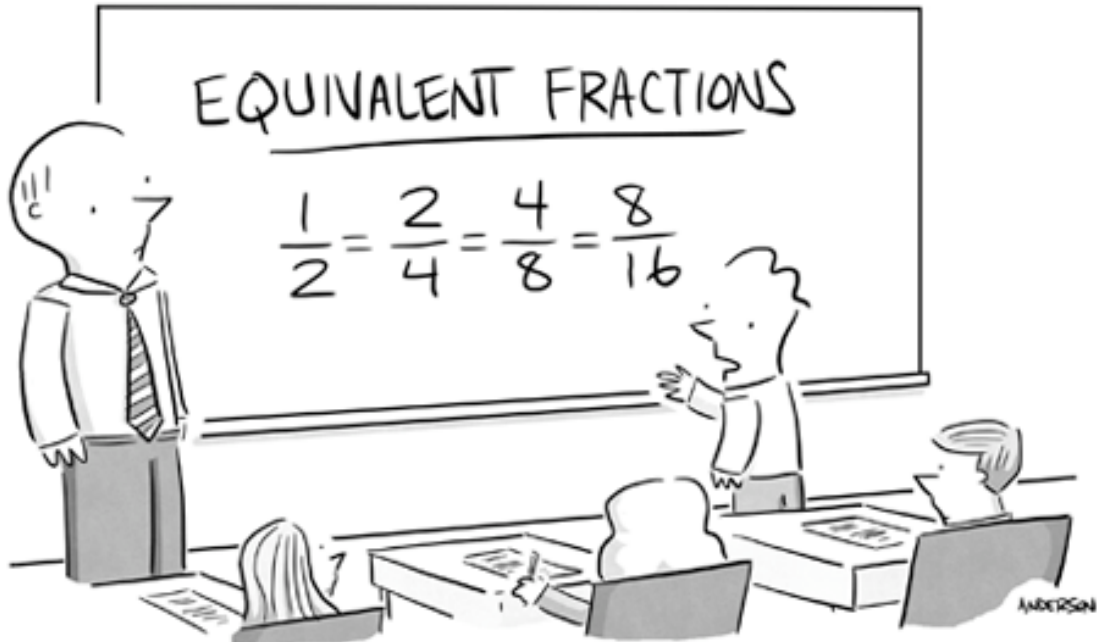
Rational Expressions

WORKBOOK

MCR3U

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"I understand they all have the same value,
but I have to tell you, the ones on the right
feel like more bang for your buck."

Chapter 2(part 1) Workbook Checklist

Worksheet	Check ✓
Intro to Rational Expressions – Fractions and Exponents Review – Worksheet	
3.3 Rational Exponents – Worksheet	
2.1/2.2 Multiplying and Dividing Rational Expressions – Worksheet	
Extra Practice Multiplying and Dividing Rational Expressions	
2.1/2.2 Adding and Subtracting Rational Expressions – Worksheet	
Rational Expressions Practice	
Review	
Practice Test	

Intro to Rational Expressions – Fractions and Exponents Review – Worksheet

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1) Add or subtract the following fractions.

a) $\frac{2}{3} + \frac{3}{4}$

b) $\frac{3}{5} - \frac{1}{3}$

c) $\frac{x}{3} + \frac{y}{2}$

d) $\frac{3x}{5} - \frac{2y}{7}$

2) Simplify (multiplication)

a) $\left(\frac{2}{3}\right)\left(-\frac{1}{5}\right)$

b) $\left(\frac{4}{5}\right)\left(\frac{25}{12}\right)$

c) $\left(\frac{17}{8}\right)\left(\sqrt{\frac{13}{289}}\right)$

d) $\left(-\frac{5}{8}\right)\left(\sqrt{\frac{75}{128}}\right)$

3) Simplify (multiplication and division)

a) $\frac{4}{9} \cdot \frac{7}{4}$

b) $\frac{-11}{5} \cdot \frac{-7}{4}$

c) $\frac{-17}{10} \div \frac{9}{4}$

d) $\frac{6}{7} \div \frac{7}{6}$

e) $\frac{7}{3} \cdot 6$

4) Simplify (exponents)

a) $(x^6)(x^4)$

b) $(y^3)(y^7)(y^9)$

c) $m^5 \div m^4$

d) $h^6 \div h^8$

e) $\sqrt{h^6} \div \sqrt{h^8}$

f) $\frac{x^{10}}{x^5}$

g) $(yz^2)^3$

h) $\left[-(x)^2\right]^2$

i) $(x^2)^3(y^3)^2$

j) $\frac{(-x)^2}{(-x)}$

5) Simplify fractions and exponents

a) $\frac{7x}{3y} \cdot \frac{12y}{5x^2} + \frac{2}{3}$

b) $\frac{25x^3}{3x} \cdot \frac{7y^2}{5yx^2}$

6) Simplify. Your answer should contain only positive exponents.

a) $(x^{-2}x^{-3})^4$

b) $(x^4)^{-3} \cdot 2x^4$

c) $(n^3)^3 \cdot 2n^{-1}$

d) $(2v)^2 \cdot 2v^2$

e) $\frac{2x^2y^4 \cdot 4x^2y^4 \cdot 3x}{3x^{-3}y^2}$

f) $\frac{2y^3 \cdot 3xy^3}{3x^2y^4}$

g) $\frac{x^3y^3 \cdot x^3}{4x^2}$

h) $\frac{3x^2y^2}{2x^{-1} \cdot 4yx^2}$

i) $\frac{x}{(2x^0)^2}$

j) $\frac{2m^{-4}}{(2m^{-4})^3}$

3.3 Rational Exponents – Worksheet

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1) Evaluate each cube root.

a) $\sqrt[3]{64}$

b) $(-1000)^{\frac{1}{3}}$

c) $\sqrt[3]{\frac{1}{8}}$

d) $\left(\frac{8}{27}\right)^{\frac{1}{3}}$

2) Evaluate each root.

a) $81^{\frac{1}{4}}$

b) $\sqrt[4]{\frac{16}{625}}$

c) $64^{\frac{1}{6}}$

d) $\sqrt[5]{-100\,000}$

3) Evaluate.

a) $8^{\frac{2}{3}}$

b) $32^{\frac{4}{5}}$

c) $(-64)^{\frac{5}{3}}$

d) $\left(\frac{1}{10\,000}\right)^{\frac{3}{4}}$

4) Evaluate.

a) $16^{-\frac{1}{4}}$

b) $25^{-\frac{3}{2}}$

c) $\left(\frac{1}{8}\right)^{-\frac{7}{3}}$

d) $\left(-\frac{1}{32}\right)^{-\frac{2}{5}}$

e) $\left(\frac{10\,000}{81}\right)^{-\frac{3}{4}}$

f) $\left(-\frac{8}{27}\right)^{-\frac{2}{3}}$

5) Simplify. Express your answers using only positive exponents.

a) $x^{\frac{1}{4}} \cdot x^{\frac{1}{4}}$

b) $\left(m^{\frac{1}{3}}\right)\left(m^{\frac{3}{4}}\right)$

c) $\frac{w^{\frac{1}{2}}}{\frac{1}{w^{\frac{2}{3}}}}$

d) $\frac{ab^2}{\frac{1}{a^2b^3}}$

e) $\left(y^{\frac{1}{2}}\right)^{\frac{2}{3}}$

f) $\left(u^{\frac{3}{4}}v^{\frac{1}{2}}\right)^{\frac{2}{9}}$

6) Simplify. Express your answers using only positive exponents.

a) $k^{\frac{3}{4}} \div k^{-\frac{1}{4}}$

b) $\frac{p^{-\frac{2}{3}}}{p^{\frac{5}{6}}}$

c) $\left(y^{\frac{2}{3}}\right)^{-3}$

d) $\left(w^{-\frac{8}{9}}\right)^{-\frac{3}{4}}$

e) $(8x)^{\frac{2}{3}}(27x)^{-\frac{1}{3}}$

f) $5\left(7y^{-\frac{2}{3}}\right)^{-2}$

7) The surface area, S , of a sphere can be expressed in terms of its volume, V , using the formula

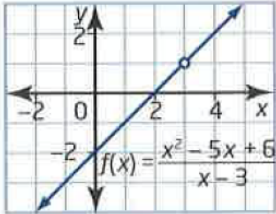
$S(V) = (4\pi)^{\frac{1}{3}}(3V)^{\frac{2}{3}}$. A beach ball has a volume of 24 000 cm³. Find its surface area, to the nearest hundred square centimeters.

2.1/2.2 Multiplying and Dividing Rational Expressions – Worksheet

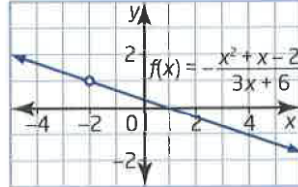
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1) State the restrictions for each function.

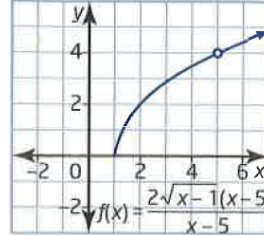
a)



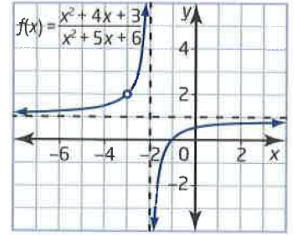
b)



c)



d)



2) Simplify each expression and state all restrictions on x .

a) $\frac{x-8}{x^2-13x+40}$

b) $\frac{x^2-3x-18}{x^2+x-42}$

c) $\frac{x+8}{x^2+6x-16}$

3) Simplify and state the restrictions on the variables.

a) $\frac{14y}{11x} \times \frac{121y}{7x}$

b) $\frac{15b^3}{4b} \times \frac{20b}{30b^2}$

c) $\frac{5x}{9y} \div \frac{5x}{18y^2}$

d) $\frac{26ab}{4a} \div \frac{39a^4b^3}{12b^4}$

4) Simplify and state the restrictions on the variable.

a) $\frac{25}{x+10} \times \frac{x+10}{5}$

b) $\frac{x+5}{x-3} \times \frac{x-3}{x+7}$

$$\text{c)} \frac{x+1}{x} \div \frac{x+1}{2x}$$

$$\text{d)} \frac{x+12}{x+10} \div \frac{x+12}{x-5}$$

5) Simplify and state the restrictions on the variable.

$$\text{a)} \frac{3x^2}{12x^2+18x} \times \frac{4x+6}{3x+30}$$

$$\text{b)} \frac{4x+24}{x^2+8x} \times \frac{12x^2}{3x+18}$$

$$\text{c)} \frac{x^2+10x+21}{x+3} \times \frac{x+2}{x^2+9x+14}$$

$$\text{d)} \frac{x^2+2x-15}{x^2-9x+18} \times \frac{x-6}{x+5}$$

6) Simplify and state the restrictions on the variable

$$\text{a)} \frac{x^2+15x}{4x+24} \div \frac{3x}{3x+18}$$

$$\text{b)} \frac{6x}{8x-72} \div \frac{9x}{2x-18}$$

$$\text{c)} \frac{x^2+15x+26}{6x^2} \div \frac{x^2-3x-10}{30x^3}$$

$$\text{d)} \frac{x^2+11x+24}{x^2+2x-3} \div \frac{x-8}{x-1}$$

7) Simplify and state the restrictions on the variable

$$\text{a)} \frac{a^2-25}{a+2} \cdot \frac{a^2-4}{a^2-7a+10}$$

$$\text{b)} \frac{y^2-4y-21}{3y^2+6y} \cdot \frac{y^2+8y}{y^2+11y+24}$$

$$\text{c)} \frac{p^2-2p+1}{p+1} \div \frac{p^2-1}{p+1}$$

$$\text{d)} \frac{x^2+6x-27}{x^2+11x+18} \div \frac{x-3}{x^2+x-2}$$

Extra Practice Multiplying and Dividing Rational Expressions

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1) Multiply the following rational expressions. Simplify and state restrictions.

a) $\frac{x+4}{x} \cdot \frac{x^2}{x^2+5x+4}$

b) $\frac{x^2+10x+16}{5x-10} \cdot \frac{x-2}{x^2+9x+8}$

c) $\frac{2x^2-10x}{x^2-9x+20} \cdot \frac{x^2-8x+16}{4x^2}$

d) $\frac{2x+4}{x+4} \cdot \frac{5x^2+21x+4}{10x+2}$

2) Divide the following rational expressions. Simplify and state restrictions.

a) $\frac{x^2-5x+6}{5} \div \frac{x-3}{15}$

b) $\frac{x^2-5x+6}{8x^2+24x} \div \frac{x-2}{4x+12}$

c) $\frac{x^2-2x+1}{x+1} \div \frac{x^2-1}{x+1}$

d) $\frac{x^2+7x+12}{x^2+3x-10} \div \frac{x^2-x-20}{x^2-25}$

2.1/2.2 Adding and Subtracting Rational Expressions – Worksheet

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1) Simplify and state any restrictions

a) $\frac{x+1}{18} + \frac{x-1}{45}$

b) $\frac{2}{3x} - \frac{1}{4x}$

c) $\frac{3}{ab} + \frac{5}{4b}$

d) $\frac{2+a}{a^2b} + \frac{4-a}{3ab^2}$

2) Simplify and state the restrictions.

a) $\frac{1}{x-6} - \frac{1}{x+6}$

b) $\frac{12}{x+8} + \frac{3}{x-9}$

$$\text{c)} \frac{x+10}{x-6} - \frac{x-3}{x+4}$$

$$\text{d)} \frac{x+5}{x+1} + \frac{x+2}{x-2}$$

3) Simplify and state the restrictions.

$$\text{a)} \frac{x}{x^2-9x+8} + \frac{2}{x-8}$$

$$\text{b)} \frac{x+3}{x+5} + \frac{x+2}{x^2+3x-10}$$

$$\text{c)} \frac{x}{x^2+3x+2} - \frac{3x-2}{x^2+8x+7}$$

$$\text{d)} \frac{x+4}{x^2-121} - \frac{2x-1}{x^2+8x-33}$$

4) Binomial expressions can differ by a factor of -1. Factor -1 from one of the denominators to identify the common denominator. Then, simplify each expression and state the restrictions.

a) $\frac{1}{x-2} - \frac{1}{2-x}$

b) $\frac{2b+3}{4b-1} + \frac{b+6}{1-4b}$

Rational Expressions Practice

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1. Simplify and state the restrictions on the variable.

a) $\frac{2x^3}{3x^2 - 4x} \times \frac{6x - 8}{2x - 2}$

b) $\frac{2x^2 - 3x}{x^2} \times \frac{x^2 - 3x - 4}{2x^2 - x - 3}$

c) $\frac{2x^2 - 13x + 20}{x} \times \frac{x^2 + 4x - 5}{x^2 - 5x + 4}$

d) $\frac{x^2 - 1}{x^2 - 4} \times \frac{x^2 - 3x + 4}{x^2 + 5x + 4}$

2. Simplify and state the restrictions on the variable.

a) $\frac{x^2 - 4}{x + 3} \div \frac{x^2 - x - 6}{x^2 + x - 6}$

b) $\frac{x^2 + 2x - 8}{x^2 + x - 6} \div \frac{x^2 + 3x - 4}{x^2 - 9}$

c) $\frac{x^2 + x}{3x^2 - 14x + 8} \div \frac{x^2 + 3x}{3x^2 - 10x - 8}$

d) $\frac{x^3 + 4x^2 + 3x}{2x^2 - 5x - 12} \div \frac{x^2 + 3x}{2x^2 + 9x + 9}$

3. Simplify and state the restrictions on the variable.

a) $\frac{4}{x^2 - x - 6} + \frac{2}{x^2 - 2x - 3}$

b) $\frac{2x}{x^2 + x - 12} - \frac{x}{x^2 - 9}$

c) $\frac{3x - 1}{2x^2 - 7x - 4} + \frac{2x - 3}{2x^2 + 3x + 1}$

d) $\frac{2x - 1}{x^2 + 3x + 2} - \frac{2x + 2}{x^2 + 5x + 6}$

4. Remove a common factor of -1 from one of the denominators and then simplify the expressions. Be sure to state any restrictions for the variable.

a) $\frac{4}{2x - 1} + \frac{2}{1 - 2x}$

b) $\frac{3x}{x - 7} - \frac{x}{7 - x}$

c) $\frac{6x - 1}{3x - 4} + \frac{4x - 1}{4 - 3x}$

d) $\frac{5x + 2}{5x - 2} - \frac{3x}{2 - 5x}$